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Institute of Health Policy, Management and Evaluation
UNIVERSITY OF TORONTO

Dalla Lana
School of Public Health

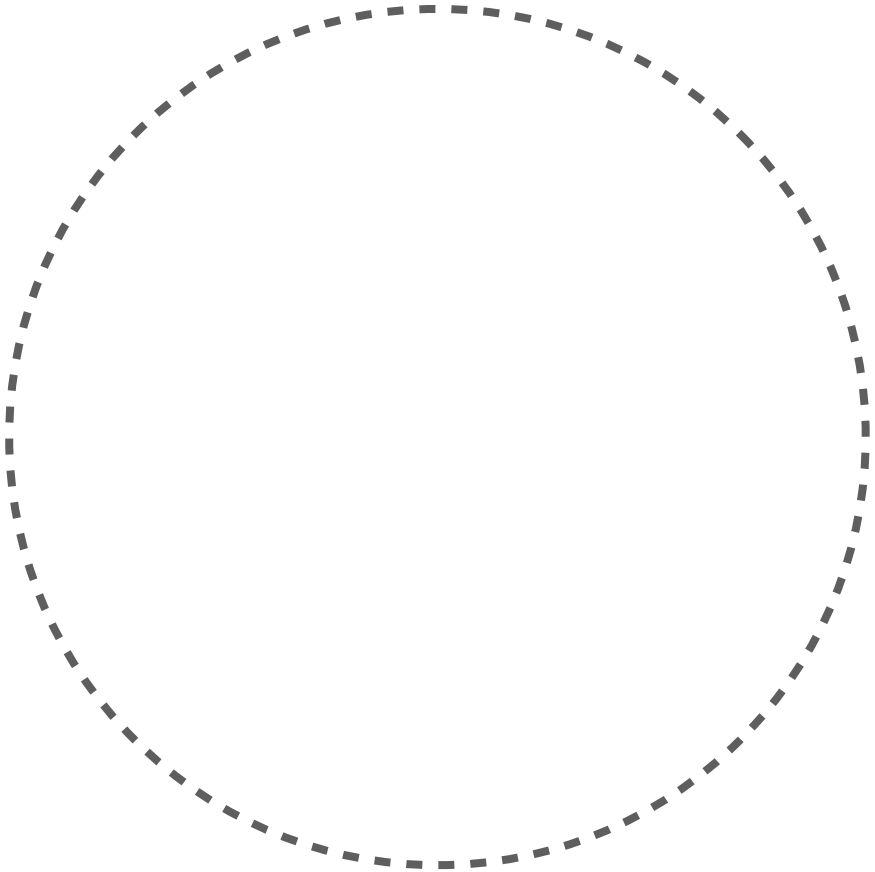


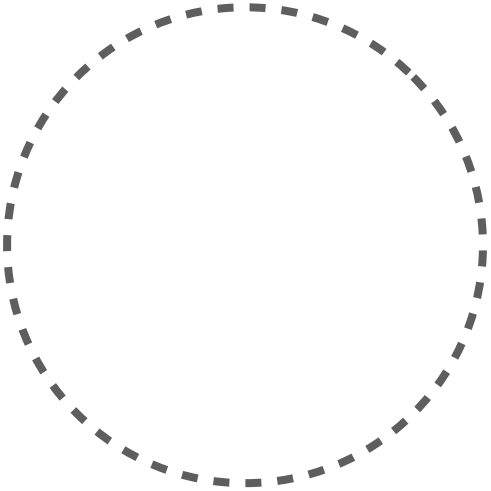
PARAMETRIC-METRIC

Example

Scenario: Imagine we have a dataset of different countries. For each country, we have information on GDP, education level, access to clean water, and vaccination rates. We also have the average life expectancy for each country.

Goal: For a new country (or a region within a country) with known GDP, education level, access to clean water, and vaccination rates, we want to predict the life expectancy (Continuous or Discrete).































- GDP: \$40,000
- Education Level (Average Years of Schooling): 10 years
- Access to Clean Water: 90%
- Vaccination Rate: 80%



Life Expectancy

 High

 Average

 Low

$$K=3$$

K=3

Life Expectancy = Average

$$K = 7$$

K=7

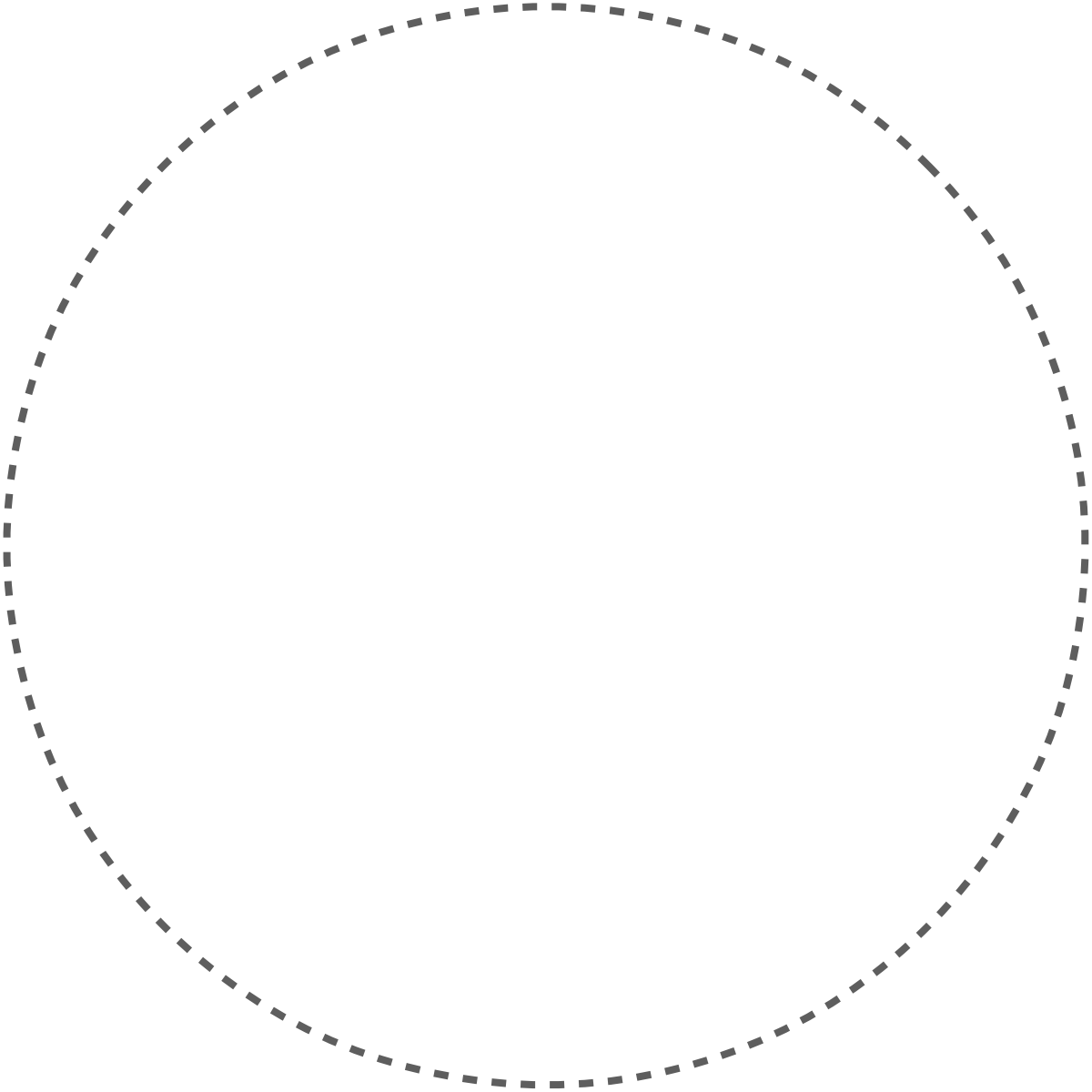
Life Expectancy = High













K=17

K=17

Life Expectancy = Low

Non-Parametric Method

PARAMETRIC VS NON-PARAMETRIC MODELS

Example

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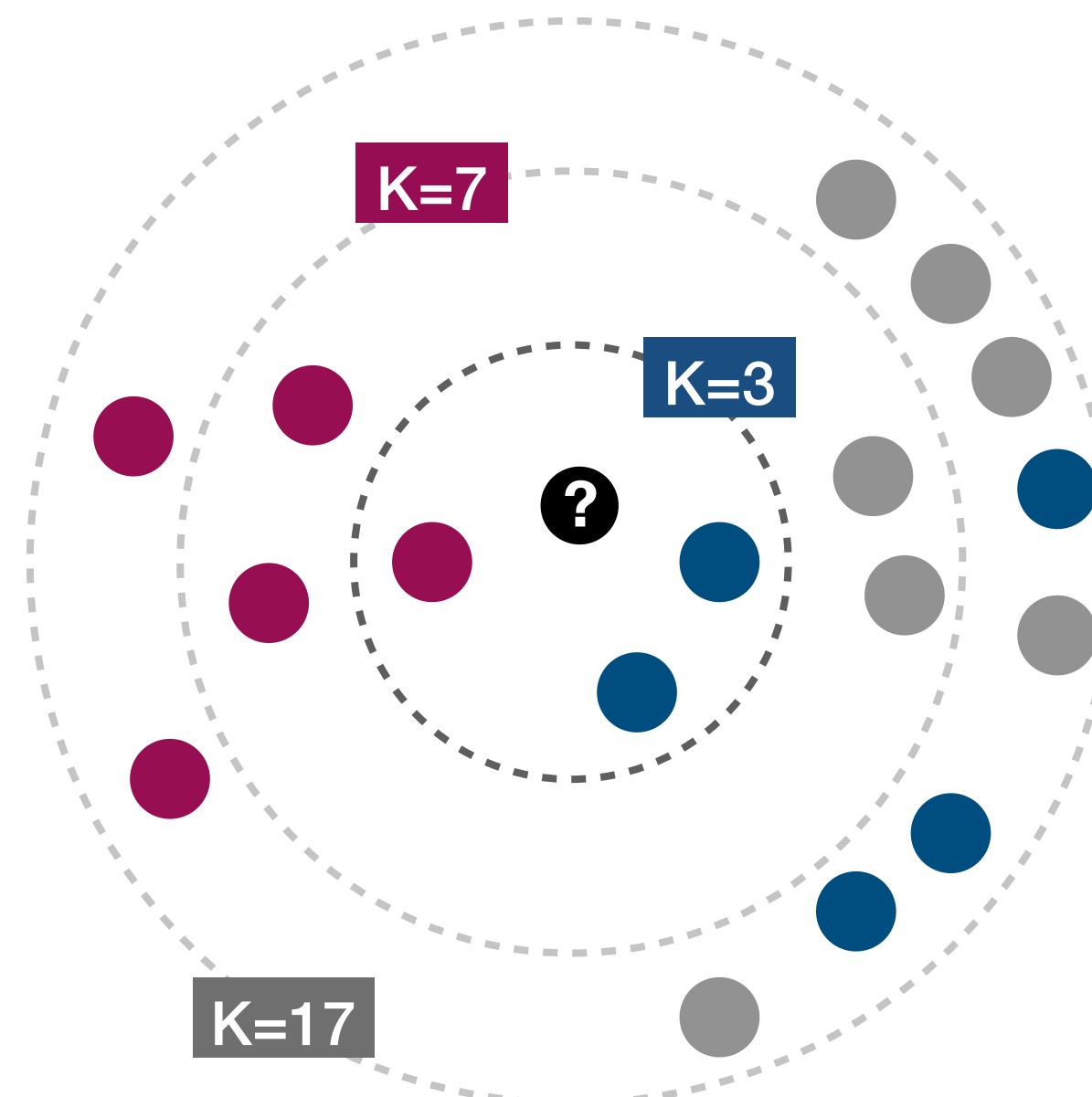
- GDP: \$40,000
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- Access to Clean Water: 90%
- Vaccination Rate: 80%

K=3 Life Expectancy = **Average**

K=7 Life Expectancy = **High**

K=17 Life Expectancy = **Low**

Non-Parametric Method



Life Expectancy

- High
- Average
- Low

PARAMETRIC OR NON-PARAMETRIC?

Parametric

Non-Parametric